

UNIT-II

Environment Pollution and Control-II

Air Pollution:

Types of air pollutants, source, effects, sink & control of common air pollutants (CO, oxides of nitrogen & sulfur, hydrocarbons and particulates), Photochemical smog, acid rain, greenhouse effect, global warming, Carbon dioxide sequestration and the concept of Carbon Credits.

AIR POLLUTION

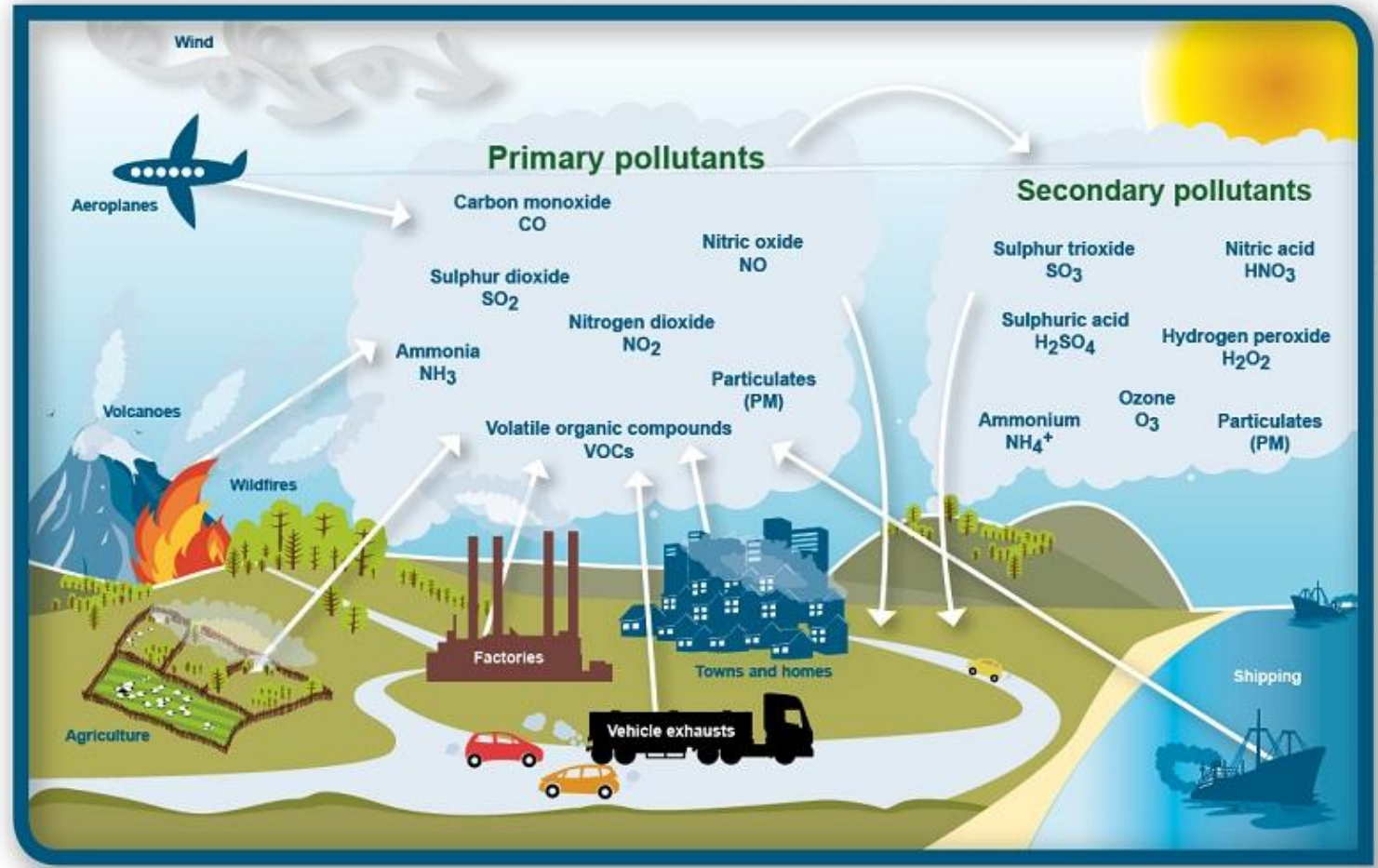
Air pollution is the addition of gaseous, liquid, or solid chemicals/substances in the air in quantities and duration that are harmful to human health and the environment. It occurs when the concentration of certain substances become high enough to cause the atmospheric environment to become toxic.

Every human being inhales about 20 kg of air daily.



Sources of air pollution

- NATURAL
- ANTHROPOGENIC



Natural sources

- Volcanic eruptions emit toxic gases and particulate matter.
- Wildfires/forest fires (add carbon monoxide)
- Microbial decay releases methane and other gases.
- Emissions from Marshy lands
- High temperatures increase the amounts of contaminants volatilizing from polluted soil and water.



Anthropogenic sources

- Deforestation activities
- Combustion activities
- Vehicular Emission
- Emission from Industries
- Bio-mass burning
- Use of pesticides, insecticides in agriculture



- **Ceremonial burnings:**

Frequent ceremonial burnings increase carbon levels in the atmosphere

- **Firecrackers:**

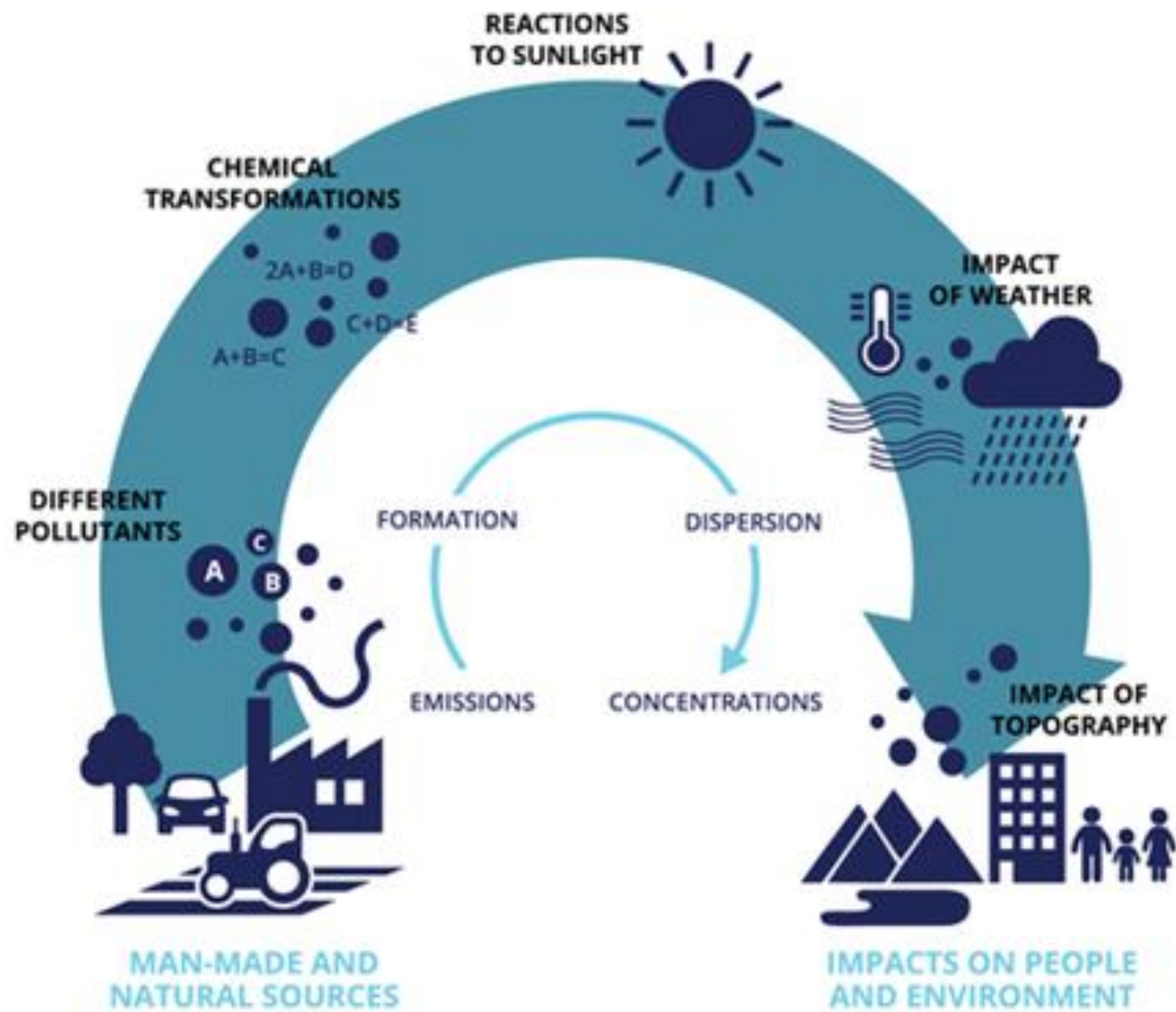
Firecrackers emit toxic substances in the atmosphere which are not measured in routine pollution checks like mercury, lead and aluminum.

Legal Burning Activities



- Fires set for recreational or ceremonial purposes (camp fires, bonfires)
- Small fires set for comfort heat at construction sites (only when air temp. is below 50°)
- Fires set for cooking (camp fires)





Types of Air Pollutants

- **PRIMARY:** Products of natural events (like fires, volcanic eruptions etc) and human activities added directly to air. Eg., CO, NO_x, SO_x, particulates and hydrocarbons.
- **SECONDARY:** Formed by interaction of primary pollutants with each other or with normal components of air. Eg., ozone, photochemical smog etc.

1. Carbon Monoxide

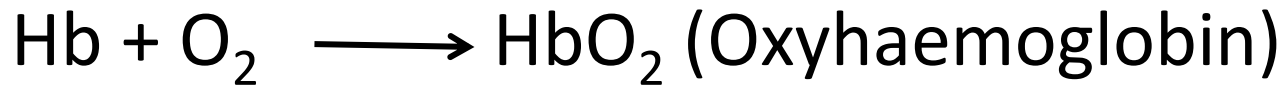
SOURCES:

- Incomplete combustion of fuels
- Automobile exhausts (accounts to ~70% of CO in atmosphere)
- Industrial operations
- Forest fires
- Volcanic eruptions
- Marsh gas emissions

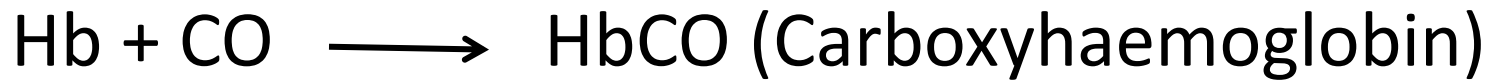
SINKS:

- Micro-organisms in soil convert CO to CO₂

EFFECTS: CO interferes with oxygen carrying capacity of blood to carry oxygen to different parts of body.



If large amount of CO is present in surroundings is inhaled, it combines with haemoglobin to form



Symptoms-laziness, exhaustion, headache. Acute shortage of oxygen can lead to coma and death (asphyxiation).

Permissible Concentrations in ambient air: 40 ppm for an exposure of 6-8 hours.

Congested Red lights: more than 100 ppm

Cigarette smoke: 400 ppm

CONTROL: Transportation sector accounts for 70-80% global emissions, specially gasoline-fed internal combustion engines

- **Modification of Engine design:** Right proportion of oxygen to be maintained for complete oxidation of C and H to CO_2 and H_2O respectively.
- **Fuel modification:** Fuels which release lesser amounts of CO to be used eg., Natural gas, methane and blends of light hydrocarbons.
- **Treatment of exhaust gases:** Two stage catalytic convertors to be used. In first stage, NO_x are reduced to N_2 and NH_3 in presence of catalyst Pt, Pd etc in presence of reducing gas such as CO. In second stage, catalysts of noble metals Pt, Ru etc. in presence of air which provides oxidising environment, supported on ceramic materials are used, which ensure oxidation of CO to CO_2 . (nitrogen does not oxidise because of low temp.)
- **Use of pollution free power:** solar and wind energy, Bio-fuels

2. Oxides of Nitrogen (NO_x)

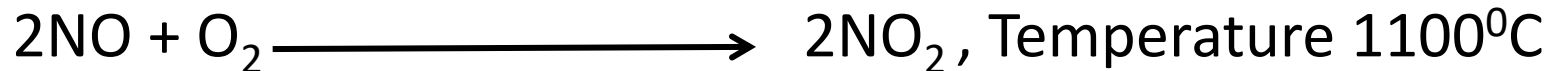
- Number of oxides of Nitrogen are introduced in atmosphere: NO, N₂O, NO₂, N₂O₃, out of these NO & NO₂ are responsible for pollution, represented as NO_x.
- NO₂ is reddish brown in colour with pungent smell and is suffocating, NO is colourless and odourless gas.

SOURCES:

Natural: During lightening discharge, N₂ and O₂ in air combine to form NO

- NO_x is also formed by the fixation of N from atmosphere or from nitrogen compounds present in organic matter (Bacterial decay of organic matter)

Man-made sources: Combustion of fossil fuels



Though first reaction is reversible but rapid cooling of combustion product prevents the dissociation of NO. Reaction 2 is very slow and formation of NO₂ is only 0.5%

- Main reaction for formation of NO₂ is reaction with ozone



SINK:

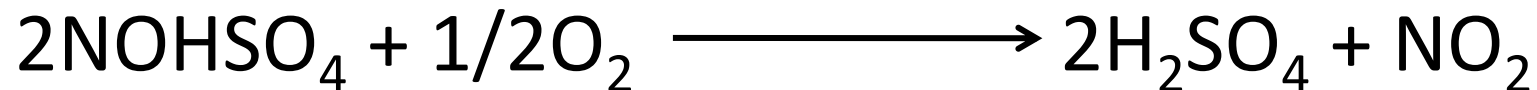
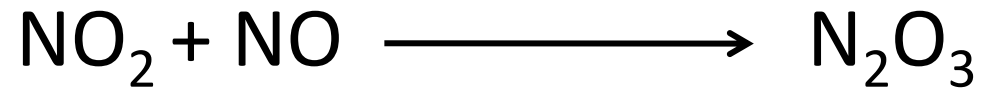
- NO and NO₂ undergo various **photochemical and chemical reactions** in atmosphere leading to formation of HNO₃, which gets precipitated during rainfall. **Ozone plays significant role in these reactions.**
- $\text{NO} + \text{O}_3 \longrightarrow \text{NO}_2 + \text{O}_2$, NO removes ozone naturally in the atmosphere.
- $\text{NO}_2 + \text{Sunlight (hv)} \longrightarrow \text{NO} + \text{O}$, Nascent oxygen reacts with oxygen molecule to form ozone. Thus **destruction and regeneration of NO₂ is a cyclic process.**
- In presence of VOCs, NO₂ reacts with water in presence of oxygen to form nitric acid $4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O} \longrightarrow 4\text{HNO}_3$
- $2\text{NO}_2 + \text{O}_3 \longrightarrow \text{N}_2\text{O}_5 + \text{O}_2$, $\text{N}_2\text{O}_5 + \text{H}_2\text{O} \longrightarrow 2\text{HNO}_3$

EFFECTS

- Almost all NO_x emissions are in form of NO, which has no known adverse health effects at concentrations found in atmosphere
- NO can oxidise to NO₂, which may react with hydrocarbons in presence of sunlight to form **photochemical smog**, which is injurious to health.
- HNO₃ formed by NO₂ causes acid rain corroding marble and metallic structures, decreasing the pH of soil and **affecting its fertility**.
- It is an acute irritant at higher concentrations than normally found. **Prolonged exposure** to relatively low concentration, has found to cause **bronchitis in children**.

CONTROL OF NO_x

- **Modifying the engine design:** Burners are so modified that fuel and air mix more slowly reducing the intensity and temperature of combustion.
- **Scrubbing the flue gases:** with H₂SO₄



Excess NO and NO₂ are circulated through the scrubber.

- **Selective catalytic reduction:** Two stage, same as discussed for CO

3. Oxides of Sulphur (SO_x)

- SO₂ and SO₃, colourless gases with pungent smell, represented as SO_x. SO₂ is second most important contributor to air pollution.

SOURCES

- Natural sources such as volcanic eruptions constitute to about 67% of SO_x pollution.
- Man made sources the sulphur emissions from petroleum refining, copper smelting and cement manufacture plants.



- The burning of fossil fuels in thermal power plants, electric power plants are also one of the main sources of SO_x emissions.



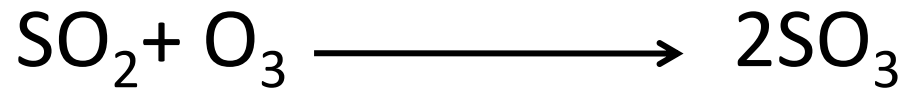
SO_x produced (from the fuel) from natural sources are uniformly distributed all over the globe but SO_x made from man-made sources is more in urban areas.

- A small amount of SO₂ undergoes photolytic oxidation in the air to form SO₃



SINKS

- The sulphur dioxide released in the atmosphere is converted to SO_3 by photolytic and catalytic processes involving ozone



- Sulphur oxides combine with water vapour to produce H_2SO_4 aerosol which give rise to “acid-rain” droplets



- H_2SO_4 combines with lime-stone minerals and thus removed from atmosphere. Lime-stone minerals act as a sink for SO_2

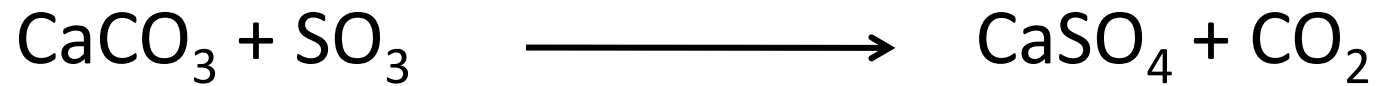
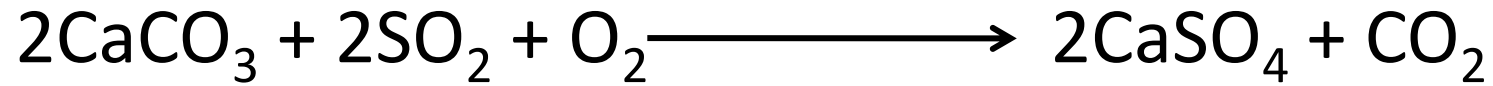
EFFECTS

- SO_2 and SO_3 are strong irritants to the respiratory tract. SO_2 is water soluble, thus is absorbed in the upper respiratory tract.
- If alongwith SO_2 , particulate matter in air is also present, the combination becomes detrimental to human health. Could cause **chronic bronchitis**.
- Plants exposed to SO_2 even at low concentration (0.03 ppm) for a few weeks, the formation of chlorophyll is slowed down resulting in loss of green colour of leaf, It is called **chlorosis**.
- Prolonged exposure causes serious damage to buildings, marble, lime stone. $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{CaSO}_4 + \text{CO}_2 + \text{H}_2\text{O}$
- CaSO_4 being water soluble washes away, leaving pitted surface.

CONTROL OF SO_x

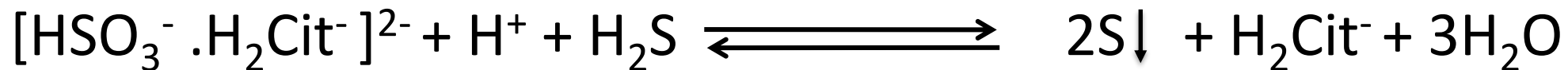
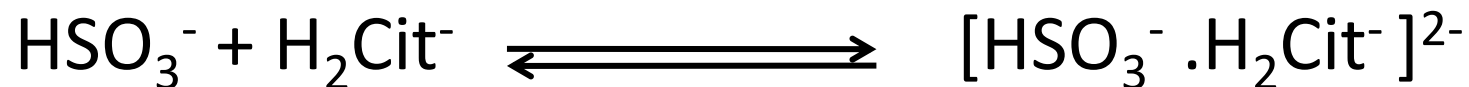
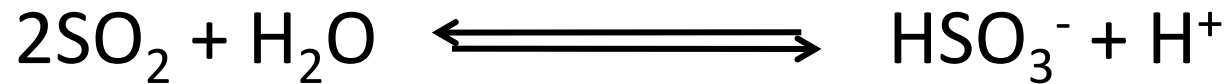
- Removal of S from fuel (coal) before burning
- Use of fuels having low content of sulphur
- By removing SO₂ from exhaust gases by using chemical scrubbers

(i) Lime Stone:



- The flue gases are passed through the slurry of CaCO₃ which reacts with SO₂ and SO₃ to form CaSO₄ (disposal is a problem)

(ii) Citric acid:



- H₂S is passed through solution containing citrate ion and Sulphur is precipitated out and citrate ions are regenerated to be used again.

4. Particulates

- It is the dispersed matter, solid or liquid in air.
- Important characteristics of particulates are:
 - **Size:** ranges from diameter of $0.002\text{ }\mu\text{m}$ to $500\text{ }\mu\text{m}$.
 - **Concentration:** from few hundred per cm^3 in clean air to 10^5 per cm^3 in polluted air.

SOURCES

Natural: volcanic eruption, wind and dust storms etc

Man-made: burning of wood, coal, oil and gaseous fuels, industrial processes, smelting and mining operations.

COMPOSITION: It could be either organic or inorganic.

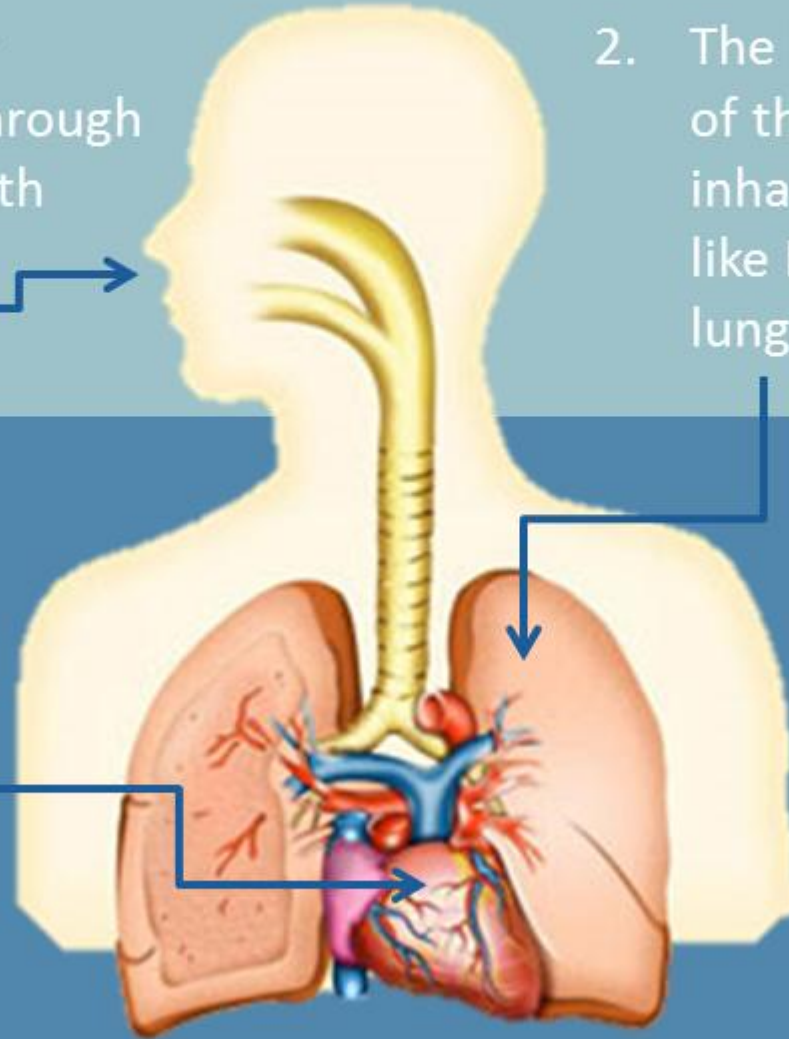
- **Organic particulates** arise from **automobile exhausts, combustion of fuels**. Organic particulates contain polycyclic aromatic hydrocarbons (PAH) which are carcinogenic.
- **Inorganic particulates** include iron oxide, calcium oxide from **combustion of coal and metallurgical operations**, Lead halides from **automobile exhausts**, asbestos dust from **industrial processes**.
- **Aerosol mists** are formed from oxidation of SO_2 to SO_3 and to form H_2SO_4 in presence of water. Cd, As, Hg also contribute to particulate matter

HOW PARTICULATE MATTER ENTERS THE BODY

1. Particulate matter enters the body through the nose and mouth when we breathe.

2. The body eliminates most of the larger particles we inhale. Smaller particles like PM_{2.5} continue to the lungs.

3. PM_{2.5} can penetrate deep into the lungs, having serious health consequences for the lungs and heart.



Harmful effects on materials:

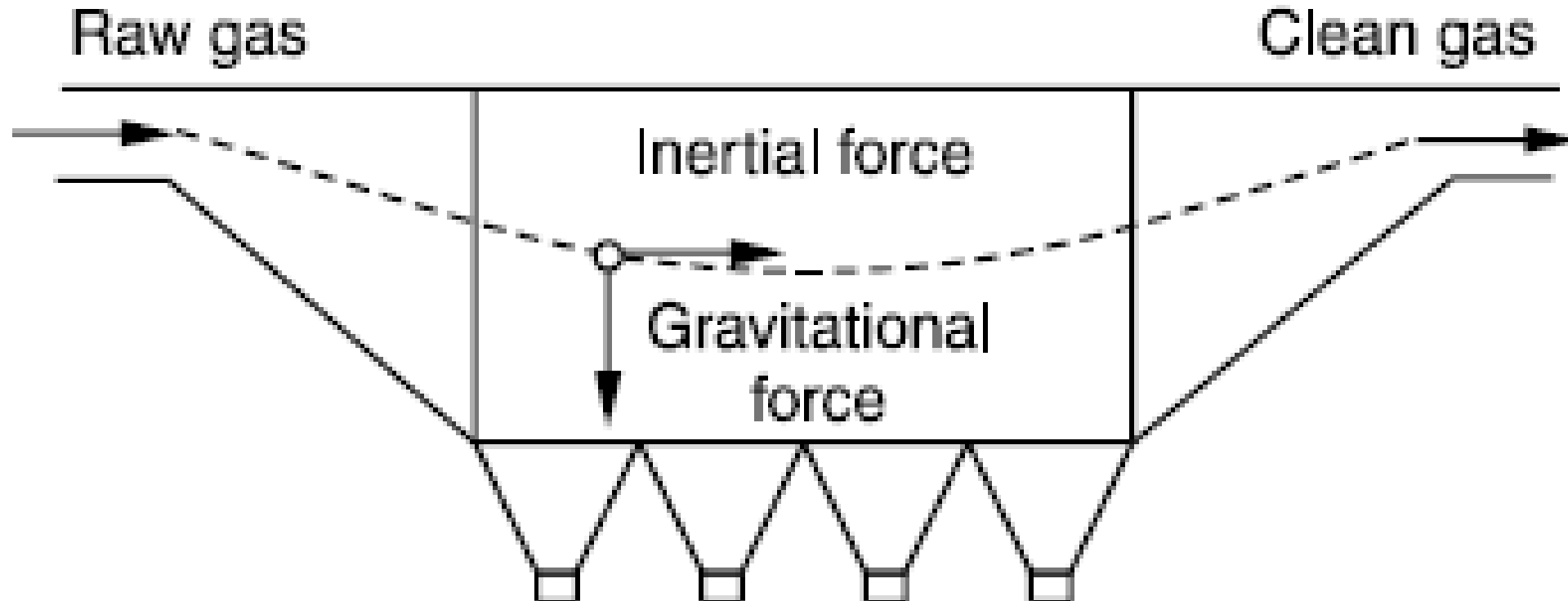
- Fumes, dust, soot, mists and aerosols can damage soil, buildings, sculpture and monuments
- Corrosive particulates accelerates corrosion of the materials
- Smoke particles stick on every surface of material, forming a film of soot

Effects on climate:

- Maintain radiation balance and heat balance of earth.
- Provide nuclei for condensation of water vapour
- Absorb free radicals, thereby acting as inhibitors for free radical chain reactions.

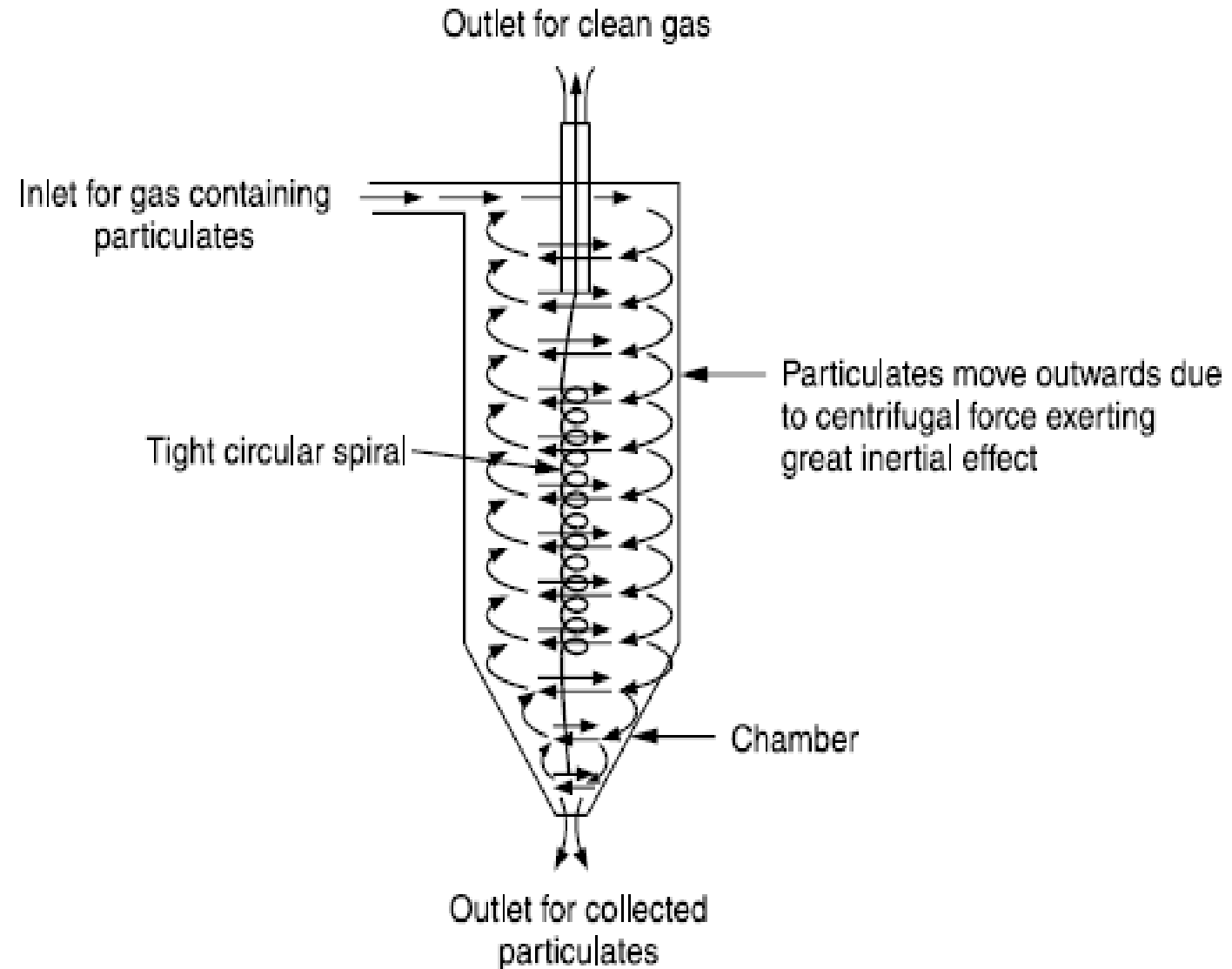
CONTROL OF PARTICULATE EMISSIONS

- **Gravity Settling Chamber:** Removes large particulates ($> 50\ \mu\text{m}$) from the gaseous effluents. Velocity of the horizontal carrier gas is reduced adequately so that particles settle by gravitational force.



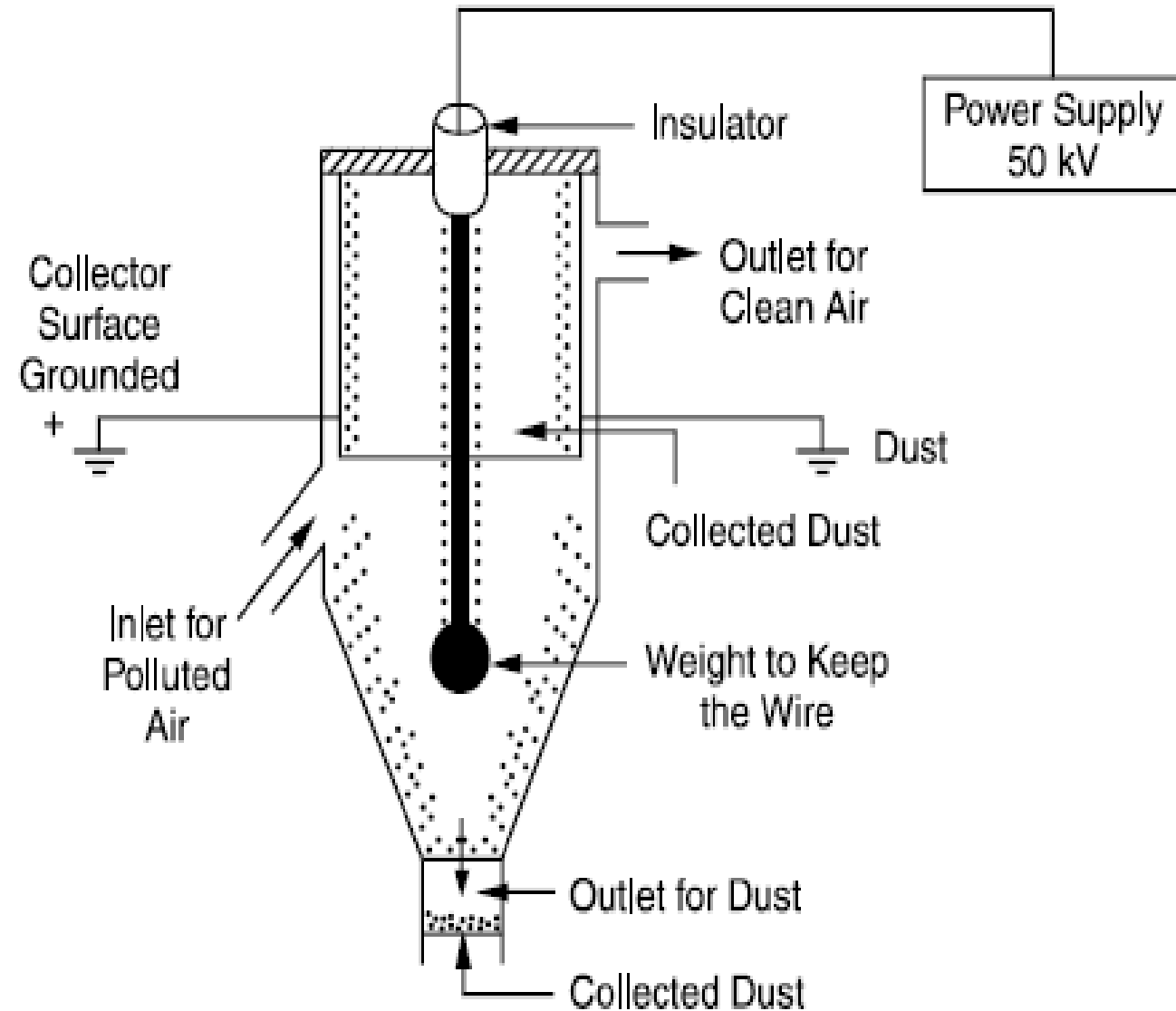
Cyclonic Separators: Gas containing particulates is allowed to flow into a tight circular spiral-fitted chamber.

Now **particulates possess greater inertia than the gas molecules and experience greater centrifugal force.** This drive these particulates towards the wall of chamber. From there they settle down due to force of gravity. Efficient for removal of large particles (5-20 μm). More efficient because centrifugal force is much greater than force of gravity.



Electrostatic precipitators:

- Used in power plants, paper and pulp industries, chemical industries, iron and steel plants.
- Consists of series of grounded thin plates charged to high voltages alternately + and -. Particles approaching these plates (around 50,000 V) become charged and are attracted towards opposite plates. From these plates they fall into a collector. Efficiency of precipitator is very high, it is capable of cleaning 150,000 litres of gas per minute at 600°C .



Wet Scrubbers:

- Exhaust gases are passed through a fine spray of water so that effective size of particle is increased due to the formation of water particulate agglomerates. These are then separated from gas stream by any collection mechanism(concept of wet precipitation)

Fabric Filters:

- Exhaust gases are forced through bags of very fine cloth-like mesh , capable of retaining small particles. Used in industries dealing with rock products, pigments etc.

5. Hydrocarbons

Sources:

- **Natural sources-** Methane is naturally occurring hydrocarbon emitted in atmosphere produced by anaerobic decomposition of organic matter in the soil, water and sediments.
- Organic matter $\longrightarrow$ $\text{CH}_4 + \text{CO}_2$ in presence of bacteria
- **Man-made sources** include Automobile exhausts, evaporation of organic solvents
- **Rice cultivation-** Paddy fields emit methane in the atmosphere. India emits 4 Tg methane in the atmosphere.
- **Ruminants** belch out lot of CH_4 daily.

SINKS: Hydrocarbons, being thermodynamically unstable tend to get oxidized in the atmosphere by a series of chemical and photochemical reactions. This gives rise to the formation of various end products such as CO₂, solid organic particulates and water soluble acids. The solid organic particulates settle down and water soluble acids are washed down by rain.

EFFECTS

- At high concentrations hydrocarbons have carcinogenic effect on lungs.
- Inhalation of benzene, toluene etc. cause irritation in the mucus membrane.
- They are responsible for the formation of photochemical smog
- Methane causes narcotic effect on human beings.
- Acetylene and propylene at 50 – 500 ppm show toxicity towards plants, damaging their growth.
- Benzopyrine present in tobacco, charcoal etc. induces cancer.

CONTROL

- **Treatment of exhaust gases:** Two stage catalytic convertors to be used.
- The hydrocarbons from vehicular emissions are controlled by the techniques such as **incineration, adsorption and absorption.**
- **Adsorption** is carried on a bed of carbon or by passing exhaust gases through a liquid in which hydrocarbons will dissolve or become suspended. **Incineration** of 'after burning' completes the oxidation of hydrocarbons to CO_2 and H_2O .
- Loss of Hydrocarbons by evaporation from fuel tanks and carburettor is reduced by **installation of a connection system** which eventually returns them to the fuel induction system.